



Republic of the Philippines

POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

OFFICE OF THE VICE PRESIDENT FOR BRANCHES AND SATELLITE CAMPUSES

STO. TOMAS BRANCH

**COMP 010– INFORMATION MANAGEMENT
FINAL PROJECT**

COMP 010– INFORMATION MANAGEMENT FINAL PROJECT

Database Title: db_tonbits

Individual: Carl Anthony L. Peña

Instructor: Mrs. Lhanie Castillo

Date: June 16, 2026



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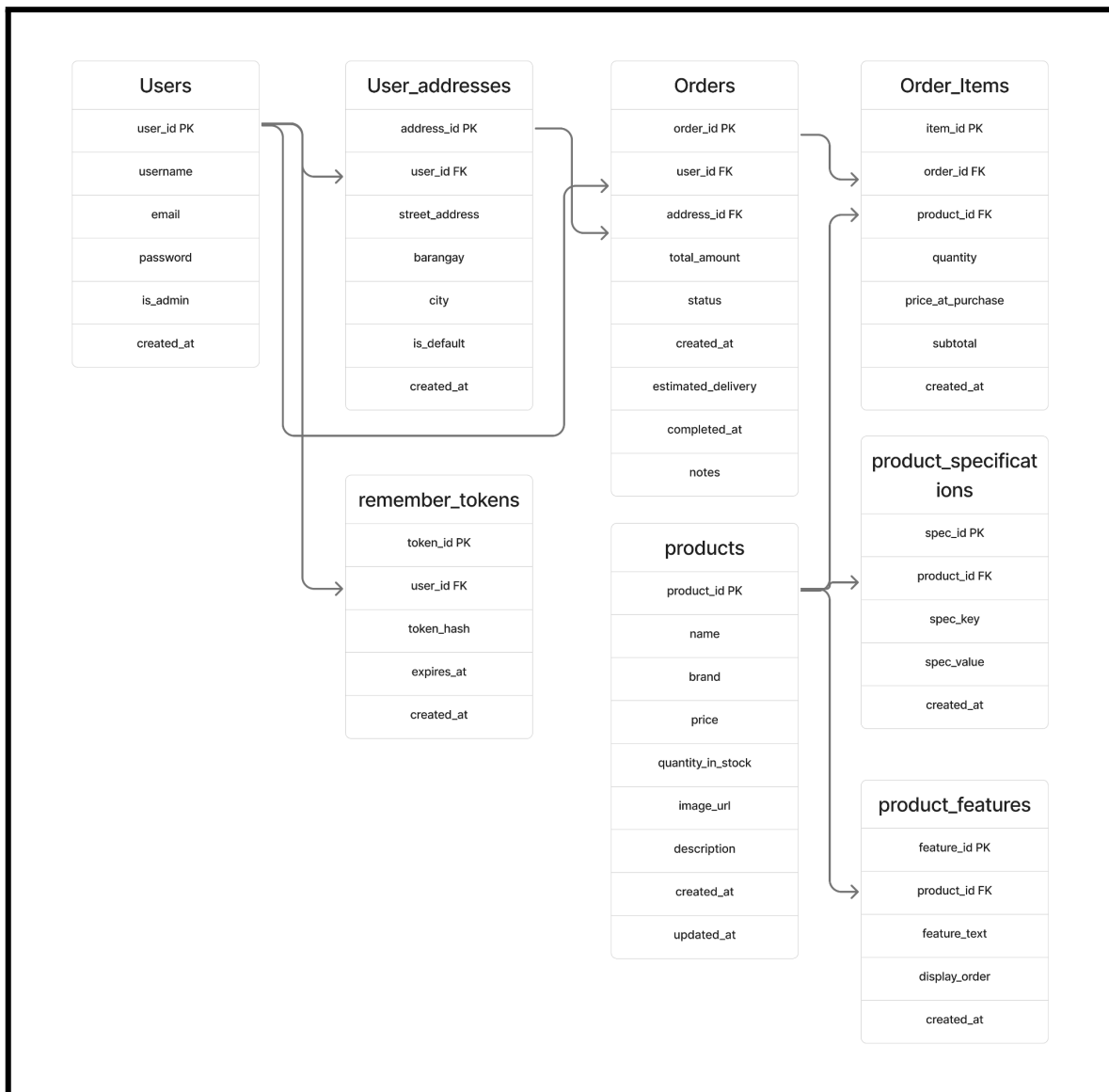
Project Description

TonBits uses a relational database with 8 interconnected tables to manage users, products, orders, and transactions while maintaining data integrity and consistency.



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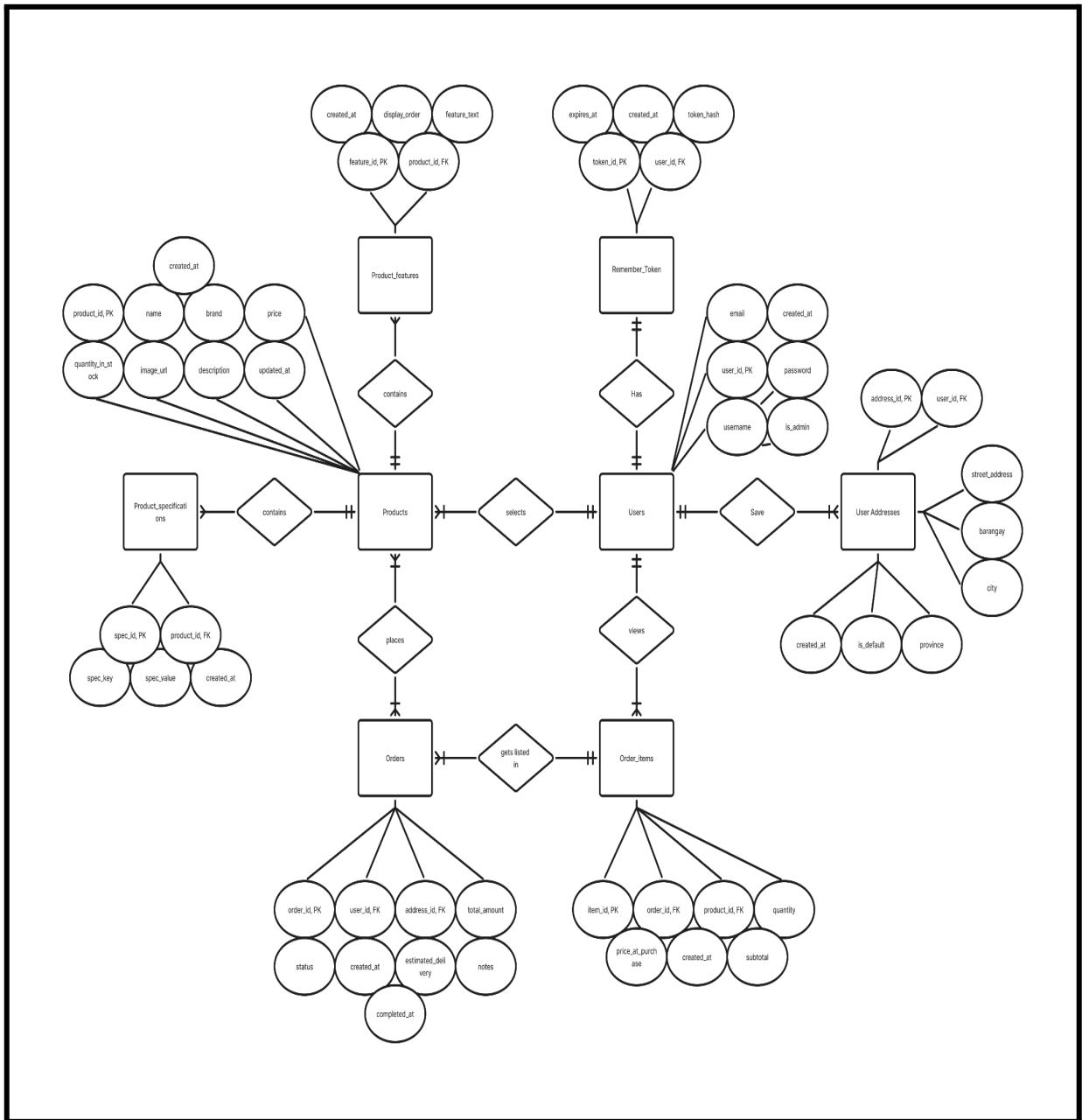
Database Schema





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Entity Relations Diagram





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Normalization Process

UNF:

Products

product_id	name	brand	price	quantity	image	specifications	features
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Orders

order_id	user_id	total_amount	status	shipping_address
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Users

user_id	username	email	password	street_address	city	province
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1NF: Products

product_id	name	brand	price	quantity_in_stock	image_url	description	created_at
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Products_Specifications

spec_id	product_id	name	spec_key	spec_value	created_at
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Products_features

feature_id	product_id	name	feature_text	display_order	created_at
------------	------------	------	--------------	---------------	------------

Users

user_id	username	email	password	street_address	city	province
---------	----------	-------	----------	----------------	------	----------

Orders

order_id	user_id	street_address	barangay	city	province	status	estimated_delivery	notes
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Order_Items

item_id	order_id	product_id	quantity	product_price	price_at_purchase	subtotal	created_at
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2NF: Products

product_id	name	brand	price	quantity_in_stock	image_url	description	created_at
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Products_Specifications

spec_id	product_id	name	spec_key	spec_value	created_at
---------	------------	------	----------	------------	------------

Products_features

feature_id	product_id	name	feature_text	display_order	created_at
------------	------------	------	--------------	---------------	------------

Users

user_id	username	email	password	street_address	city	province
---------	----------	-------	----------	----------------	------	----------

Orders

order_id	user_id	street_address	barangay	city	province	status	estimated_delivery	notes
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Order_Items

item_id	order_id	product_id	quantity	price_at_purchase
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3NF:

Products

product_id	name	brand	price	quantity_in_stock	image_url	description	created_at
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Products_Specifications

spec_id	product_id	name	spec_key	spec_value	created_at
---------	------------	------	----------	------------	------------

Products_features

feature_id	product_id	name	feature_text	display_order	created_at
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Users

user_id	username	email	password	created_at	is_admin
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User_Addresses

address_id	user_id	street_address	barangay	city	province	is_default	created_at
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Orders

order_id	user_id	address_id	status	created_at	estimated_delivery	completed_at	notes	payment_method
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Order_Items

item_id	order_id	product_id	quantity	price_at_purchase	subtotal	created_at
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Data Dictionary

Table 1: Users

Column	Data Type	Constraint	Sample Data	Description
user_id	INT	PK, AUTO_INCREMENT	1	Unique user identifier
username	VARCHAR(20)	NOT NULL, UNIQUE	johndoe	User login name
email	VARCHAR(255)	NOT NULL, UNIQUE	john@example.com	User email address
password	VARCHAR(255)	NOT NULL	\$2y\$10\$zE...	Hashed password
is_admin	BOOLEAN	DEFAULT FALSE	FALSE	Admin privilege flag
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	2026-06-09 13:23:00	Account creation date



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Table 2: User_Address

Column	Data Type	Constraint	Sample Data	Description
address_id	INT	PK, AUTO_INCREMENT	12	Unique address identifier
user_id	INT	FK (users), NOT NULL	1	Reference to user
street_address	VARCHAR(255)	NOT NULL	#10 Rizal St	Street and house number
barangay	VARCHAR(100)	NOT NULL	San Jose	Barangay name
city	VARCHAR(100)	NOT NULL	Lucena City	City name
province	VARCHAR(100)	NOT NULL	Quezon	Province/state name
is_default	BOOLEAN	DEFAULT FALSE	TRUE	Default address flag



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created_at	TIMESTAMP	DEFAULT CURRENT_TIMEST AMP	2026-06-09 13:23:00	Address creation date
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Table 3: Products

Column	Data Type	Constraint	Sample Data	Description
product_id	INT	PK, AUTO_INCREMENT	101	Unique product identifier
name	VARCHAR(255)	NOT NULL	GeForce RTX 4090	Product name
brand	VARCHAR(100)	NOT NULL, INDEX	NVIDIA	Brand name
price	DECIMAL(10,2)	NOT NULL	112000.00	Product price in PHP
quantity_in_stock	INT	DEFAULT 0	15	Available stock quantity
image_url	VARCHAR(255)	-	/images/rtx4090.png	Path to product image



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description	TEXT	-	24GB GDDR6X...	Product description
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTA MP	2026-06-09 13:23:00	Date product added
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTA MP	2026-06-09 13:23:00	Last update timestamp



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Table 4: Products_Specifications

Column	Data Type	Constraint	Sample Data	Description
spec_id	INT	PK, AUTO_INCREMENT	501	Unique specification ID
product_id	INT	FK (products), NOT NULL	101	Reference to product
spec_key	VARCHAR(100)	NOT NULL, INDEX	Memory	Specification name (e.g., Memory)
spec_value	VARCHAR(255)	NOT NULL	24GB GDDR6 X	Specification value (e.g., 24GB)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	2026-06-09 13:23:00	Date specification added



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Table 5: Products_Features

Column	Data Type	Constraint	Sample Data	Description
feature_id	INT	PK, AUTO_INCREMENT	901	Unique feature ID
product_id	INT	FK (products), NOT NULL	101	Reference to product
feature_text	VARCHAR(255)	NOT NULL	DLSS 3.0 Support	Feature description
display_order	INT	DEFAULT 0	1	Display order in list
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	2026-06-09 13:23:00	Date feature added



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Table 6: Orders

Column	Data Type	Constraint	Sample Data	Description
order_id	INT	PK, AUTO_INCREMENT	5001	Unique order identifier
user_id	INT	FK (users), NOT NULL	1	Reference to customer
address_id	INT	FK (user_addresses), NULLABLE	12	Reference to shipping address
total_amount	DECIMAL(10,2)	NOT NULL	112000.00	Order total in PHP
status	ENUM	DEFAULT 'pending', INDEX	pending	Order status



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created_at	TIMESTAMP	DEFAULT CURRENT_TIME STAMP	2026-06-09 13:23:00	Order placement date
estimated_ delivery	DATETIME	-	2026-06-15 14:00:00	Expected delivery date
notes	TEXT	-	Leave at front door	Special order notes



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Table 7: Order_Items

Column	Data Type	Constraint	Sample Data	Description
item_id	INT	PK, AUTO_INCREMENT	7001	Unique line item ID
order_id	INT	FK (orders), NOT NULL	5001	Reference to order
product_id	INT	FK (products), NOT NULL	101	Reference to product
quantity	INT	NOT NULL	1	Quantity ordered
price_at_purchase	DECIMAL(10, 2)	NOT NULL	112000.0 0	Price at time of order
subtotal	DECIMAL(10, 2)	NOT NULL	112000.0 0	Item total (price × quantity)



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created_at	TIMESTAMP	DEFAULT CURRENT_TIMEST AMP	2026-06-09 13:23:00	Item creation date
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Table 8: Remember_Tokens

Column	Data Type	Constraint	Sample Data	Description
token_id	INT	PK, AUTO_INCREMENT	8001	Unique token ID
user_id	INT	FK (users), NOT NULL	1	Reference to user
token_hash	VARCHAR(255)	NOT NULL, UNIQUE, INDEX	a8f9c2d3..	Hashed token value
expires_at	DATETIME	NOT NULL, INDEX	2026-07-09 13:23:00	Token expiration timestamp
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	2026-06-09 13:23:00	Token creation date

